

The exact latent-sum P-OPD gate switches the teacher off

Why $T = 1$ does not train, what causes it, and the temperature that follows

measured on probe run w10kj2v8 (FLUX.2-klein 9B $\rightarrow$ 4B)

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Abstract

P-OPD gates teacher matching by the posterior responsibility of the teacher component in a two-component Gaussian mixture over the observed transition. At the documented default `popd_temperature`= 1.0, the responsibility is the *exact* joint density ratio summed over all latent dimensions. This note records that this setting does not train: measured on a 9B $\rightarrow$ 4B probe, 75.4% of transitions fall below $\gamma = 0.01$, the median responsibility is 8×10^{-12} , and `grad_norm` sits at 2.2×10^{-5} with a loss of 9.9×10^{-7} . The cause is not a bad teacher and not a bug — the implementation checks out to five decimals — but dimension aggregation: the joint KL obeys $K = \frac{D}{2}w^2$ with $D = 131072$, so a per-dimension mismatch of $w = 0.4\%$ of one noise standard deviation already half-closes the gate, and the measured w reaches 9.8%. A second observation is that K varies by 1.2×10^4 along the denoising axis while varying only $\approx 30\%$ between samples at fixed step, so at any single global temperature the gate is primarily a weighting along the trajectory, decomposing into $226\times$ from scheduler geometry and $53\times$ from a genuine growth of the teacher–student velocity gap. The temperature deployed for the long run follows from the measurement: $T = \bar{K} = 107$ with $\alpha = 0.731$.

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1 The gate under test

Let x_t be the on-policy state reached by the behavior student, and let the sampler take a stochastic step with a model-independent scalar covariance $\sigma_{\text{tr}}^2 I$ (Flow-SDE/Dance-SDE: $\sigma_{\text{tr}}^2 =$

`std_dev_t2 | $\Delta t$ |`; CPS: $\sigma_{\text{tr}}^2 = \text{std_dev_t}^2$). Three transition means live at that state: the behavior mean μ_{old} that actually produced the observed next latent y , the teacher mean μ_T , and the trainable student mean μ_θ . P-OPD forms

$$\log \rho = \log \frac{\mathcal{N}(y \mid \mu_T, \sigma_{\text{tr}}^2 I)}{\mathcal{N}(y \mid \mu_{\text{old}}, \sigma_{\text{tr}}^2 I)} = \sum_{i=1}^D \frac{(y_i - \mu_{\text{old},i})^2 - (y_i - \mu_{T,i})^2}{2\sigma_{\text{tr}}^2}, \quad (1)$$

$$\gamma = \sigma \left(\text{logit } \alpha + \frac{\log \rho}{T} \right), \quad \mathcal{L} = \mathbb{E}_{\text{batch}} \left[\text{sg}(\gamma) \cdot \frac{1}{D} \sum_{i=1}^D \frac{(\mu_{\theta,i} - \mu_{T,i})^2}{2\sigma_{\text{tr}}^2} \right], \quad (2)$$

where D is the latent event dimension, $\alpha \in (0, 1)$ is the prior probability of the teacher component, $T > 0$ is the temperature, and `sg` is a stop-gradient. Code: `compute_popd_responsibility` and `compute_popd_gaussian_mean_kl` in `src/flow_factory/trainers/xopd/common.py`, wired in `_precompute_popd_step_caches` / `_optimize_train_pass` in `src/flow_factory/trainers/xopd/trainer.py`.

The quantity that will carry the whole analysis is the joint KL between the two Gaussians,

$$K = \text{KL}(\mathcal{N}(\mu_{\text{old}}, \sigma_{\text{tr}}^2 I) \parallel \mathcal{N}(\mu_T, \sigma_{\text{tr}}^2 I)) = \frac{\|\mu_T - \mu_{\text{old}}\|^2}{2\sigma_{\text{tr}}^2}, \quad (3)$$

logged per sample as `popd/teacher_old_kl_joint`. Note that (3) is the dimension-summed version of the transition KL L_{KL} of `per_timestep_loss_dominance_theory.tex`, with the SDE transition variance σ_{tr}^2 in place of the flow-matching noise fraction; that note’s analysis of late-step dominance therefore applies directly and is picked up in §4.2.

2 The exact law of the log density ratio

Proposition 1 (Gaussian law of the ratio). *Let $y \sim \mathcal{N}(\mu_{\text{old}}, \sigma_{\text{tr}}^2 I_D)$ be the observed transition, i.e. evaluate the ratio under the policy that generated it. Then*

$$\log \rho \sim \mathcal{N}(-K, 2K), \quad (4)$$

and consequently $\mathbb{E}[\rho] = 1$, $\text{median}(\rho) = e^{-K}$, and $\Pr[\rho > 1] = \Phi(-\sqrt{K/2})$.

Proof. Write $y = \mu_{\text{old}} + \sigma_{\text{tr}} \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_D)$, and let $\Delta = \mu_T - \mu_{\text{old}}$. Then $y - \mu_T = \sigma_{\text{tr}} \varepsilon - \Delta$ and

$$\|\sigma_{\text{tr}} \varepsilon\|^2 - \|\sigma_{\text{tr}} \varepsilon - \Delta\|^2 = 2\sigma_{\text{tr}} \varepsilon^\top \Delta - \|\Delta\|^2, \quad (5)$$

so from (1), $\log \rho = \varepsilon^\top \Delta / \sigma_{\text{tr}} - \|\Delta\|^2 / (2\sigma_{\text{tr}}^2) = \varepsilon^\top \Delta / \sigma_{\text{tr}} - K$. The first term is Gaussian with mean 0 and variance $\|\Delta\|^2 / \sigma_{\text{tr}}^2 = 2K$, which gives (4). The moment identity $\mathbb{E}[\rho] = \exp(-K + \frac{1}{2} \cdot 2K) = 1$ is the log-normal mean, as it must be since $\mathbb{E}_{\pi_{\text{old}}}[\rho] = \int \pi_T = 1$. The median and tail probability follow from (4) directly. $\square$

Two consequences shape everything below.

The mean of ρ is uninformative. $\mathbb{E}[\rho] = 1$ holds for every K , however large, while the median is e^{-K} . At the measured $K \approx 100$ the median ratio is 10^{-44} and the unit mean is carried entirely by excursions of probability $\Phi(-\sqrt{K/2}) = \Phi(-7.06) \approx 8 \times 10^{-13}$. This is the familiar collapse of importance weights in high dimension; a finite batch sees $\rho \approx 0$ everywhere.

Saturation is a statement about K , not about the teacher. At $T = 1$ and $\alpha = \frac{1}{2}$, $\gamma < 0.01$ whenever $\log \rho < -4.595$, which by (4) has probability $\Phi((K - 4.595)/\sqrt{2K})$. This exceeds 0.999 already at $K \approx 30$. Whether the gate is open is therefore decided by the magnitude of one scalar, and §4 is about what sets that magnitude.

3 Measurement

Probe: `xopd_configs/sde_pathwise/flux2_klein_9b_to_4b_popd_diag_t1.yaml`, three epochs on 32 GPUs, FLUX.2-klein-base-9B teacher into a LoRA FLUX.2-klein-base-4B student, Flow-SDE at `noise_level= 0.7`, four trained transitions per rollout drawn from the 28-step grid, 512px, $\alpha = 0.5$, $T = 1$. Run `w10kj2v8`. Because the probe performs exactly one optimizer step per epoch, every reading below is taken at $\theta = \theta_{\text{old}}$, i.e. strictly on policy.

3.1 The implementation is sound

Three logged quantities are self-validating, and they pass before any conclusion is drawn from the run (Table 1). This matters: a saturated gate and a broken covariance would look alike in the loss.

Table 1: Correctness checks available from the diagnostics. The first confirms that the variance used in (1) is the variance y was actually drawn with; the second that replaying the student on the stored state reproduces the mean the rollout used; the third that the ratio obeys Proposition 1.

quantity	expected	measured	verdict
<code>popd/old_innovation_rms</code> = $\text{RMS}[(y - \mu_{\text{old}})/\sigma_{\text{tr}}]$	1	0.999993 ± 0.002	pass
<code>popd/behavior_drift_rms</code> = $\text{RMS}[(\mu_{\theta} - \mu_{\text{old}})/\sigma_{\text{tr}}]$	0 at $\theta = \theta_{\text{old}}$	0 exactly	pass
$\mathbb{E}[\log \rho]$ vs $-K$ (Prop. 1)	equal	-99.60 vs -99.53 (0.07%)	pass
rollout vs replay σ_{tr}^2 , Δt	equal	no mismatch raised	pass

Remark. *The measured $\text{sd}[\log \rho] = 129.4$ is far above the $\sqrt{2K} = 14.1$ that (4) predicts at fixed K . This is not a contradiction: the statistic pools transitions with different K , and $\text{sd}(K) = 128.6$ across the pool. The pooled spread is dominated by K varying, not by the Gaussian draw — the observation that §4.2 turns into the main structural finding.*

3.2 At $T = 1$ there is no training signal

Table 2: Global statistics over the three probe epochs at $T = 1$, $\alpha = 0.5$, $D = 131072$. The gate is closed on three quarters of transitions and never opens fully; the resulting gradient is five orders of magnitude below the clipping threshold (`max_grad_norm= 1.0`).

quantity	value	quantity	value
$\bar{K}$ (joint KL)	99.53	γ mean	0.0522
$\text{sd}(K)$	128.6	γ median	8.4×10^{-12}
K range	[1.03, 818]	γ p90 / p99	0.196 / 0.766
K/D (per dimension)	7.6×10^{-4}	$\Pr[\gamma < 0.01]$	0.754
$\mathbb{E}[\log \rho]$	-99.60	$\Pr[\gamma > 0.99]$	0
$\log \rho$ range	$[-803.6, 3.66]$	gate entropy (max 0.693)	0.092
<code>train/loss</code>	9.86×10^{-7}	<code>train/grad_norm</code>	2.22×10^{-5}

Figure 1 resolves this along the trajectory. The gate is partially open only on the earliest steps and is off from step 16 onward, where every transition in the pool falls below $\gamma = 0.01$. The measured γ sits slightly above the curve $\sigma(-\bar{K}_k)$ evaluated at the mean K of each step, as it must: γ is convex in $\log \rho$ in this regime, so averaging the gate exceeds the gate of the average.

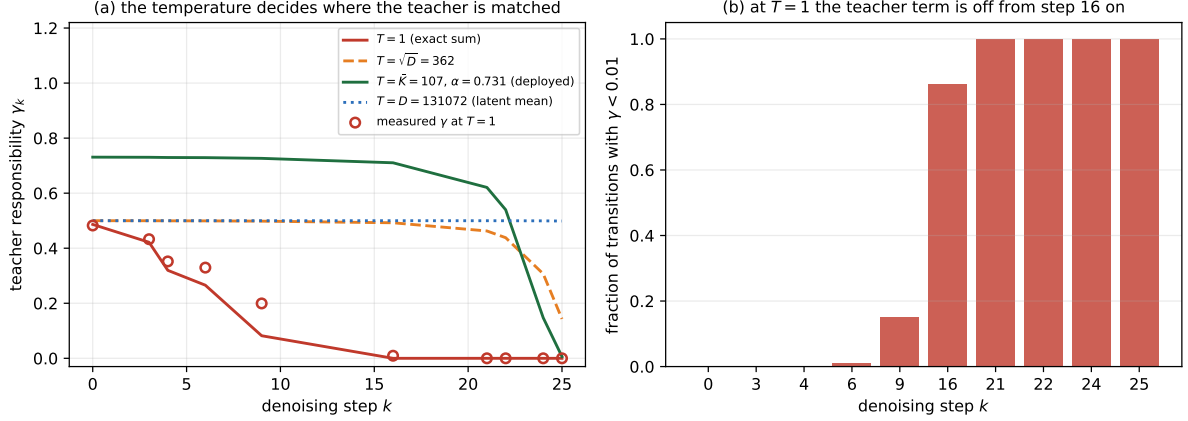


Figure 1: (a) Teacher responsibility along the denoising axis. Curves are $\sigma(\text{logit } \alpha - \bar{K}_k/T)$ at four temperatures; circles are the probe’s measured γ at $T = 1$. $T = D$ (dotted) is flat at α — a constant loss multiplier rather than a gate. (b) Fraction of transitions with $\gamma < 0.01$ at $T = 1$, per step.

4 Root cause

4.1 Dimension aggregation sets the scale of K

Define the per-dimension mismatch in units of the transition noise,

$$w = \text{RMS} \left[\frac{\mu_T - \mu_{\text{old}}}{\sigma_{\text{tr}}} \right] = \sqrt{\frac{1}{D} \sum_{i=1}^D \frac{(\mu_{T,i} - \mu_{\text{old},i})^2}{\sigma_{\text{tr}}^2}}, \quad \text{so} \quad K = \frac{D}{2} w^2. \quad (6)$$

This is logged as `popd/teacher_old_gap_whitened_rms`, and the identity reproduces the measured K per step to within 3% (Table 3, columns K and $\frac{D}{2}w^2$; the residual is the difference between the mean of a square and the square of a mean).

Equation (6) is the root cause. An open gate at $T = 1$ needs $K \lesssim 1$, hence

$$w \lesssim \sqrt{2/D} = 3.9 \times 10^{-3} \quad \text{at} \quad D = 131072. \quad (7)$$

The gate therefore demands that the teacher’s transition mean agree with the student’s to within 0.4% of one noise standard deviation *in every one of 131072 coordinates simultaneously*. Two distinct models do not satisfy this except where they happen to be nearly identical. The measured w ranges from 8.9×10^{-4} at step 0 — below the threshold, which is exactly why the gate is alive there — to 9.8×10^{-2} at step 25, a 10% per-coordinate disagreement that (6) converts into $K = 648$ (Figure 2).

Why this is not fixable by a better teacher

K grows linearly in D at fixed per-coordinate agreement. Raising the resolution from 512px to 1024px multiplies D by four and K with it, so a gate that is marginal at one resolution is dead at the next — and a *better* teacher only helps as w^2 , i.e. halving the per-coordinate mismatch buys a factor of four against a factor of D . The exact latent sum is therefore not a tunable-but-correct choice; it is a quantity whose scale is set by the latent dimension, which is what the temperature exists to undo.

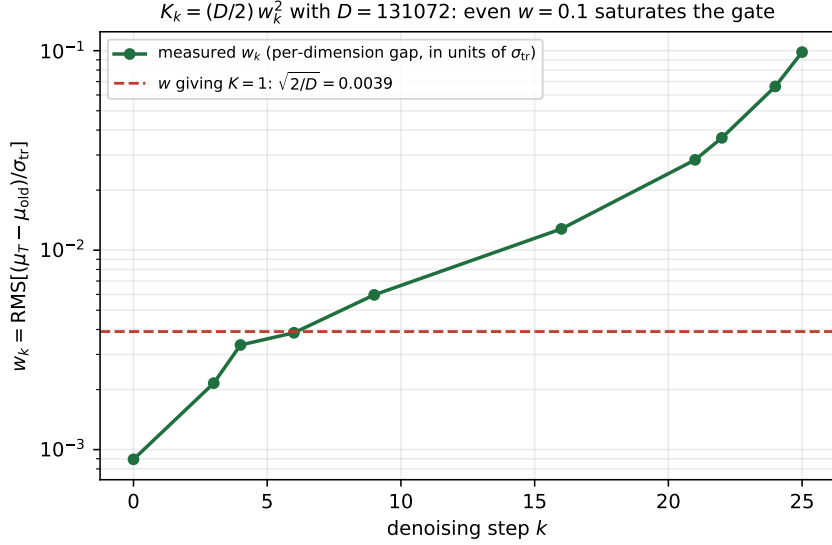


Figure 2: Per-dimension mismatch w_k against the threshold (7) at which the exact-sum gate half-closes. The measured mismatch is small in absolute terms everywhere and still crosses the threshold after step 4, because K multiplies it by $D/2$.

4.2 Where along the trajectory: scheduler geometry versus a real teacher gap

Using $\mu = x_t + v \Delta t$ and $\sigma_{\text{tr}}^2 = \text{std}^2|\Delta t|$, (3) factors as

$$K = \frac{\|v_T - v_{\text{old}}\|^2 \Delta t^2}{2 \text{std}^2|\Delta t|} = \underbrace{\frac{\|v_T - v_{\text{old}}\|^2}{2}}_{\text{teacher disagreement}} \cdot \underbrace{\frac{|\Delta t|}{\text{std}^2}}_{\text{scheduler geometry}}. \quad (8)$$

Both factors grow toward the end of the trajectory, and the measurement separates them (Table 3, Figure 3): from step 0 to step 25 the scheduler factor grows $226\times$ (the step size grows $18\times$ while the transition variance shrinks $12\times$) and the velocity disagreement grows $53\times$. Their product is the observed 1.2×10^4 .

This is the same split that `per_timestep_loss_dominance_theory.tex` draws between Cause A, a parameterization artifact from Δt^2 amplification, and Cause B, genuine difficulty near the data manifold. P-OPD inherits both through K , and the practical consequence is specific to the gate: at any single global T , the between-step variation (1.2×10^4) overwhelms the between-sample variation at fixed step ($\text{sd}(K)/K \approx 0.3$, Table 3), so

*one global temperature makes the responsibility a weighting along the denoising axis,
with only a $\pm 30\%$ per-sample modulation on top.*

That is a weaker claim than the method’s motivation suggests — the gate is advertised as suppressing transitions where the teacher is implausible, and it does, but under a single temperature “where” is mostly “how late in the trajectory”.

5 Consequence: choosing the temperature

The temperature rescales $\log \rho$ before the sigmoid, so it selects which regime of Figure 1 the run lives in. Table 4 evaluates the four candidates using the measured K , and Figure 4 shows the loss weight $\gamma_k K_k$ each step ends up receiving — the quantity that decides what the run actually optimizes.

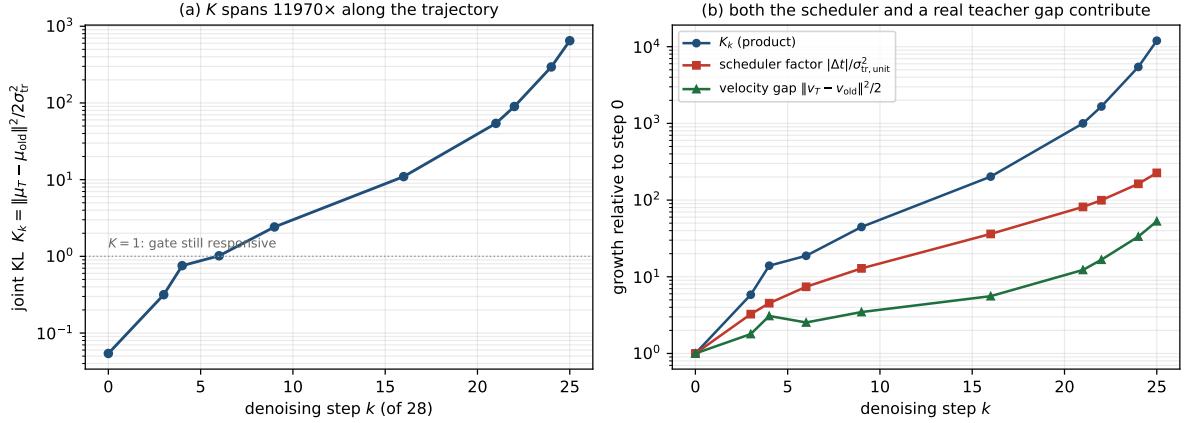


Figure 3: (a) Joint KL per trained step, spanning nearly four decades. (b) The factorization (8), each curve normalized to step 0: the scheduler contributes $226\times$ and the teacher-student velocity gap $53\times$.

Table 3: Per-step profile from the probe (means over the pool; $D = 131072$). w is the per-dimension mismatch (6) and $\frac{D}{2}w^2$ its predicted K . The last two columns are the measured responsibility at $T = 1$ and the fraction of transitions the gate closes. Only the ten steps drawn during the three probe epochs appear; the sampler redraws them each epoch.

step k	K	w	$\frac{D}{2}w^2$	$ \Delta t $	σ_{tr}^2	$ \Delta t /\text{std}^2$	γ	$\Pr[\gamma < 0.01]$
0	0.054	0.00089	0.052	0.0057	0.490	0.012	0.483	0.00
3	0.316	0.00215	0.304	0.0070	0.182	0.038	0.433	0.00
4	0.755	0.00334	0.733	0.0075	0.141	0.053	0.352	0.00
6	1.017	0.00385	0.972	0.0086	0.099	0.087	0.330	0.01
9	2.415	0.00597	2.333	0.0109	0.073	0.151	0.199	0.15
16	10.95	0.01278	10.71	0.0221	0.052	0.424	0.009	0.86
21	54.16	0.02837	52.75	0.0450	0.047	0.954	0.000	1.00
22	89.93	0.03656	87.60	0.0539	0.046	1.166	0.000	1.00
24	294.8	0.06620	287.2	0.0818	0.043	1.908	0.000	1.00
25	648.0	0.09838	634.3	0.1046	0.040	2.650	0.000	1.00

Table 4: Temperature candidates evaluated at the measured $\bar{K} = 99.5$ with $\text{sd}(K) = 128.6$ and $D = 131072$. Quantiles of γ combine the Gaussian draw of (4) with the spread of K across the pool. The last two columns describe the loss weight profile over the ten observed steps.

setting	T	α	γ p01 / p50 / p99 (model)	final-step share	weight spread
exact latent sum	1	0.5	0.000 / 0.000 / 1.000	— (no signal)	$10^{23}\times$
$\sqrt{D}$	362	0.5	0.218 / 0.427 / 0.665	36.3%	$3.4 \times 10^3\times$
calibrated	107	0.731	0.035 / 0.500 / 0.965	2.9%	$1.2 \times 10^3\times$
latent mean	131072	0.5	0.499 / 0.500 / 0.501	58.7%	$1.2 \times 10^4\times$
no gate ($\gamma \equiv \alpha$)	—	0.5	constant	58.8%	$1.2 \times 10^4\times$

The quantile column treats the pooled $\log \rho$ as one Gaussian with the measured $\bar{K}$ and $\text{sd}(K)$, so it overstates the tails: the pool is a mixture over steps whose K differs by four decades, not a single Gaussian. At $T = 1$ the model gives p50/p99 of 0.000/1.000 against the measured $8 \times 10^{-12}/0.766$. It is quoted because it makes the three regimes comparable at a glance; the exact statements are the per-step curves of Figure 1 and the weight shares of Figure 4, which use the measured per-step K without this approximation.

Read the two ends against the middle:

- $T = 1$ gives no signal, for the reason of §4.1.
- $T = D$ reproduces a per-dimension (latent-mean) log ratio. It is perfectly stable and perfectly uninformative: γ spans 0.499 to 0.501, so the run is the ungated objective scaled by α — its loss weight profile is indistinguishable from no gate at all (58.7% versus 58.8% of the weight on the final step).
- $T = \bar{K}$ places the median gate logit one unit below logit α ; choosing $\alpha = \sigma(1) = 0.731$ then centers the median responsibility at 0.5, which is the widest usable range. Transitions closer than typical to the teacher get $\gamma > 0.5$, further ones $\gamma < 0.5$.

The deployed configuration is therefore $T = 107$ (the mean K over the probe’s three epochs; the final-epoch summary alone gives 99.5) and $\alpha = 0.731$, in `xopd_configs/sde_pathwise/flux2_klein_9b_to_4`. Its effect on what gets optimized is substantial (Figure 4b): the ungated objective spends 58.8% of its teacher-matching weight on the single final step, while the calibrated gate spends 2.9% there and redistributes onto steps 21–24, which is the region where the teacher is still plausible under the student’s own transition density but the two models genuinely differ.

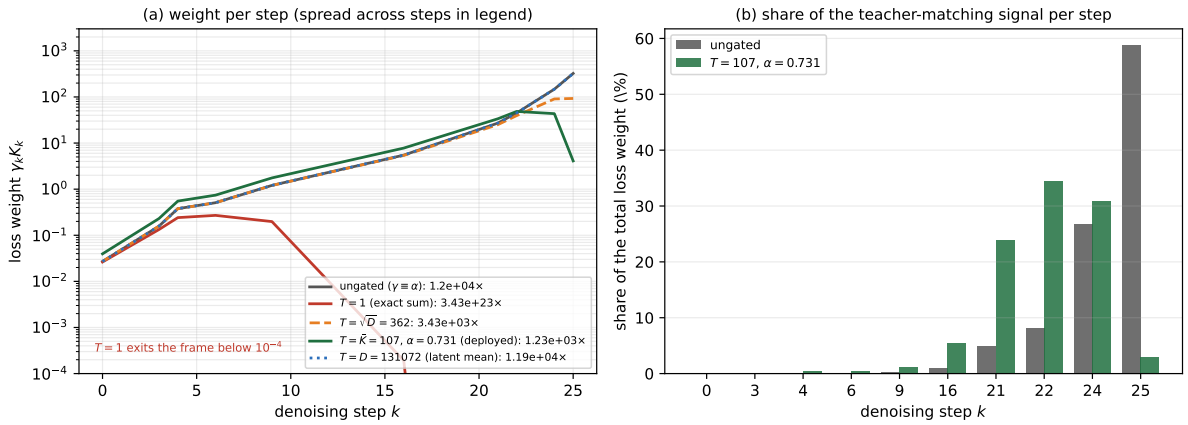


Figure 4: (a) Loss weight $\gamma_k K_k$ per step; the $T = 1$ curve leaves the frame below 10^{-4} . (b) Share of the total teacher-matching weight per step, ungated versus the deployed gate. Shares are conditional on the ten steps the probe happened to draw.

6 Open points

A per-timestep temperature would make the gate per-sample. The structural finding of §4.2 is that a global T spends its dynamic range on the trajectory axis. Setting $T_k \propto \mathbb{E}[K_k]$ — one temperature per denoising step, estimated online — would divide out the $226\times$ scheduler component and the $53\times$ systematic teacher trend, leaving the $\pm 30\%$ between-sample variation as the only driver. The gate would then mean what the method says it means: this transition is less plausible than others *at the same point in the trajectory*. This is not implemented; it is the natural next variant if the global- T arm shows the gate doing little.

The gate should open as distillation proceeds. Every number here is measured at $\theta = \theta_{\text{old}}$ with an undistilled student, so w is the base 4B-versus-9B mismatch. As training reduces w , K falls quadratically and γ rises, so at fixed T the gate self-anneals toward α . `popd/gamma_mean` and `popd/teacher_old_kl_joint` over the long run test this directly; if they do not move, the gate is not tracking distillation progress.

Attribution still needs a control. The P-OPD arm differs from the 32B→4B ODE mix baseline in the sampler, the loss space and the gate at once. Crediting any change to the gate requires a `xopd_target_mode: direct` run on the identical Flow-SDE sampler, which has not been run.

Reproducing

```
# probe (three epochs, 32 GPUs) -- the exact-sum case under test
MASTER_PORT=29620 bash scripts/xopd_cluster/run_4node_xopd.sh \
  xopd_configs/sde_pathwise/flux2_klein_9b_to_4b_popd_diag_t1.yaml

# temperature/alpha recommendation + the trajectory profile, from the probe run
python scripts/xopd_analysis/calibrate_popd_gate.py \
  --run-name flux2klein_9b_to_4b_popd_DIAG_sum_t1

# figures and the cached statistics used in this note
python scripts/xopd_analysis/plot_popd_gate_temperature.py \
  --run 315229706-xi-an-jiaotong-university-/Flow-Factory-XOPD/w10kj2v8
```

Cached probe statistics: `docs/xopd/figures/popd_gate_probe_stats.json`. Algorithm reference: `docs/opd/cross_opd_xopd.md`, section “P-OPD — probability-mixture proximal target”. Companion theory on per-step weighting: `docs/xopd/per_timestep_loss_dominance_theory.tex`.